PRIME MINISTERS' OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT MANYARA REGION
FORM FOUR MOCK EXAMININATION 2026

CODE: 033/1 **BIOLOGY**

TIME. 3:00HRS **AUGUST, 2026**

INSTRUCTIONS

1. This paper consists of three sections A, B, and C with total of eleven (11) questions.
2. Answer all questions from section A and B, and only 2 question from section C.
3. Section A carries 16 marks, section B carries 54 marks and section C carries 30 marks.
4. All writing should be in blue or black ink except for drawings which should be in pencil.
5. Communication devices and any unauthorized materials are not allowed in the examination room.
6. Write your **Examination number (index number)** on every page of your answers booklets provided

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SECTION A (16 Marks)

Answer all questions from this section

1. For each of the item (i-ix) choose the correct answer from the given alternatives and write it's latter beside the item number in the answer sheet provided.

i) Form four students attended the school biology laboratory for grinding hibiscus in order to obtain its juice for practical activity. Which set of apparatus are they going to use when grinding the eaves

A. Test tube and test tube rack
B. Microscope and hand lens
C. Beaker and Bunsen burner
D. Spatula and tongs
E. Mortar and pestle

ii) Amina has a farm around the Indian Ocean. She planted maize and germinated. After five days all maize were dried. Which conditions cause the maize to die when planted near Indian Ocean?

A. Crenation
B. Turgidity

C. Diffusion
D. Haemolysis
E. Plasmolysis

iii) Practically higher animals have tongue for a purpose in life. This tongue can either be rolled or unrolled especially when animal is performing various activities. In animal tongue rolling is an example of
A. Continuous variation
B. Swallowing
C. Lubrication food
D. Discontinuous variation
E. Mechanical digestion

iv) A plant contains root hair cell that plays a role in absorption of nutrients and water from the soil to other parts of the seed do so?
A. Plumule
B. Radical
C. Cotyledons
D. Micropyle
E. Testa

v) A form four (IV) students had a problem of dislocation in the knee. Which of the following tissues were injured seriously.
A. Cartilage
B. Ligament
C. Synovial membrane
D. Tendon
E. Muscles

vi) A blood group B patient hospitalized for blood deficiencies can only receive blood from two possible donors.
A. B^- or O^+
B. B^- or O^-
C. B^+ or AB^-
D. A^- or O^+
E. B^- or AB^+

vii) If you go to the market and buying sweet potatoes and onions you will actually be buying
A. Onion bulb and stem tuber
B. Steam tuber and flesh leaves
C. Root tubers and stem tuber
D. Root tubers and flesh leaves
E. Onion bulb and flesh leaves

viii) When first day of menstruation was 8th April on which date is a woman of 28 days cycle most likely to ovulate?
A. 26th March
B. 22th April
C. 5th May
D. 16th April
E. 21th April

ix) Maria started salivating as she saw a ripe peeled pineapple. What part of the brain is concerned with Marias Salivation
A. Hypothalamus
B. Medulla oblongata
C. Cerebellum
D. Cerebrum
E. Cerebral hemisphere

x) As a form four student how can assist a form one student on picking STI'S among the following
A. Protected sex
B. Delaying sex
C. Avoiding sharing of sharp project
D. Keeping the environment clean
E. Take a balanced diet

2. Match items in List A with the correct response in list B by writing the later of the correct response beside the item number in the answer booklet provided.

LIST A	LIST B
i) Carries oxygenated blood towards the gonads	A. Carotid artery
ii) Carries deoxygenated away from the kidney	B. Hepatic artery
iii) Carries oxygenated blood towards the stomach	C. Renal vein
iv) Carries oxygenated blood towards the head and neck	D. Renal artery
v) Carries oxygenated blood towards the liver	E. Genital artery
vi) Carries deoxygenated blood away from posterior parts and hind limbs	F. Genital vein
	G. Iliac vein
	H. Mesenteric artery
	I. Vena cava
	J. Pulmonary vein

SECTION B (54 Marks)

Answer all questions in this section

3. (a) Getrude can see well the writing on the black board when she seats at the back bench than she seats on the front seat.
   i) Give the name of eye disorder she is suffering from **(01 Marks)**
   ii) Explain how the disorder can be corrected **(01 Marks)**

   (b) Give five differences between response you exhibited when you remove your fingers from fire accidentally touched and response you exhibited by salivating after your friend graphically described your favorite delicious dish **(05 Marks)**

   (c) Briefly explain how the endocrine system help a person to be safe from danger of being attacked by the lion by quickly climbing the tree. **(02 Marks)**
4. (a) How do the following supports the theory of organic evolution
   i) Comparative anatomy **(01 Marks)**
   ii) Cell biology **(01 Marks)**
   iii) Comparative embryology **(01 Marks)**

   (b) Giving example in each case distinguish between divergent and convergent evolution **(03 Marks)**

   (c) Why is Lamarck theory of evolution no longer accepted? Three points **(03 Marks)**
5. (a) A flower was found to have the following characteristics; dull coloured petal, large anthers small pollen grain and large freely hanging stigma
   i) What is the likely agent of pollination of the flower? **(0½ Marks)**
   ii) What is the significance of large hanging stigma in plants? **(0½ Marks)**

   (b) How is meiosis promoting genetic variation in population? Explain three points **(03 Marks)**

(c) Mama Asha was told by her doctor that the fluid surround her unborn baby was essential for normal development and protection throughout the pregnancy. Using your knowledge of biology identify the fluid and give its four significance to the developing fetus **(05 Marks)**

6. (a) Differentiate between continuous variation from discontinuous variation, give three points **(03 Marks)**

   (b) Assume that in human being the gene for Brown eyes is dominant over that for blue eyes.

   i) A brown – eyed man marries a blue – eyed woman and they have eight brown eye children. What are the genotypes of all the individuals in the family **(03 Marks)**

   ii) What is the probability that the first child produced by parents who are both heterozygous for brown eyes will have blue – eyed. **(03 Marks)**

7. (a) (i) What do you understand by the term oxygen dept? **(01 Marks)**

   (ii) Write a word equation for the anaerobic respiration of glucose in muscle cells and Bean plant. **(02 marks)**

   (b) Mention three factors that enable arteries to transport blood at great distance **(03 Marks)**

   (c) Differentiate arteries and veins in terms of

   i) Valves

   ii) Lumen

   iii) Direction of blood flow

8. (a) Parents brought children to a health centre one was diagnosed with marasmus and other was diagnosed with kwashiorkor. State the cause of each of the disorder as distinct from each other and give three symptoms of marasmus that distinguish it from kwashiorkor **(05 Marks)**

   (b) How would you tabulate the differences between active immunity and passive immunity? Two points **(02 Marks)**

   (c) Explain how HIV/AIDS weakens the immune system and increase the vulnerability to opportunistic diseases **(02 Marks)**

SECTION C (30 Marks)

Answer only two questions from this section

9. Explain three ways through which plants gets rid of waste materials they produce and three reasons for why plants do not necessarily need elaborate excretory organ **(15 Marks)**

10. Some organism can fly, swim, walk and crawling while others moves only a part or parts of bodies for various biological and physiological purposes. Use your biological knowledge with examples to describe the statement.

(15 Marks)

11. The human vertebral column consist of 33 irregular bones called vertebrae. These bones are classified based on their location along the vertebral column from the neck to the tail. In light to this information describe five types of human vertebrae. **(15 Marks)**